

Daniel Iván Parra Verde

Full-Stack ML Engineer • FastAPI & React • NLP/Agents • AWS • since 2023 (+3 years)

Portfolio: danivpv.com | GitHub: [danivpv](https://github.com/danivpv) | LinkedIn: [danivpv](https://www.linkedin.com/in/danivpv) | HuggingFace: [danivpv](https://huggingface.co/danivpv)

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EDUCATION

UNAM-CFATA

2021 | BSc. TECHNOLOGY

GRONINGEN UNIVERSITY (G)

2019 | EXCHANGE (AI/CS)

Netherlands; Full Scholarship

UNAM-IIMAS

2022 | MSc. CANDIDATE IN MATH

SKILLS

AI and ML Libraries:

PyTorch • LangChain • LlamaIndex •
Hugging Face • scikit-learn • Pandas •
NumPy • Docling • Gemini • OpenAI •
Anthropic • OpenCV

Frameworks:

FastAPI • React • Next.js • Temporal •
Spark • SQLAlchemy • Django • Kafka

Cloud Services:

AWS EMR • AWS S3 • AWS ECS • AWS
SageMaker • AWS Lambda • AWS
CloudWatch • AWS API Gateway

Databases:

PostgreSQL • MongoDB • Redis •
Qdrant • SQLite

Programming Languages:

Python • SQL • Bash • TypeScript •
Terraform • Julia • Wolfram

Software Tools:

Docker • CI/CD • Git • GitHub Actions
• Sentry • Postman • $\LaTeX$ • Notion

Spoken Languages

English (+C1) • German (+A2) • Spanish

COURSEWORK

Bachelor

Deep Learning in **Python**(G)
Relational Databases in **SQL** (G)
Artificial Intelligence in **Python** (G)
Algorithms and Data Structures in **C** (G)
Linear Algebra

Master

Statistical Inference
Theoretic Machine Learning
Information Theory
Statistical Foundations of Privacy
Numerical Analysis **Python**, **Julia**
Real and Functional Analysis

WORK EXPERIENCE

SENIOR DATA SCIENTIST

Arkham Technologies | April 2025 – Present | Remote

- Engineered end-to-end NLP and LLM integrations for AI agents; integrated into **React** UI to test inference APIs via **FastAPI** to 100+/week users.
- Scaled 1K JUMEX ML pipelines on 100 **AWS** workers via **Spark**, **Temporal**, **Kafka**; designed scalable backend architecture.
- Led **CI/CD** of **Docker** containers for staging and prod ML platform via **AWS** (ECS, EMR), **GitHub Actions**, **Terraform**. 10× faster deployments vs baseline.
- Automated AI agent monitoring and NLP evaluation (LLM-as-Judge) via **FastAPI**, **PostgreSQL**, and **Temporal**.
- Debugged **FastAPI/Kafka** ML and AI microservices via **CloudWatch** and **Sentry**.

AI MENTOR & DEVELOPER

LIDR | November 2024 – April 2025 | Remote

- Deliver sessions, technical slides, live-coding sessions, and workshops on AI-assisted software development.

DATA SCIENTIST

Kuona | November 2023 – June 2024 | Remote

- Tuned **PyTorch** RNNs for sales forecasting in enterprise retail revenue management for major clients (12% forward MAPE); **AWS** batch ML via **Django**.
- Modeled competitors' pricing data, reducing forecasting MAPE by 40% and enabling price-demand elasticity analysis via **Pandas** and **scikit-learn**.
- Designed QLoRA and sales-weighted loss functions for revenue KPIs.

DATA SCIENTIST JR.

Entropia AI | February 2023 – November 2023 | Remote

- Built 8 **ETL** data processing pipelines (**Python**, **bs4**, **Airflow**) ingesting 2K docs with 100M+ tokens into **MongoDB** for legal partners.
- Engineered NLP **RAG** pipelines; raised custom QA LLM-as-Judge accuracy from 8% to 72% on legal corpora using **LlamaIndex**, **MongoDB** and **LangChain**.
- Agile** methodology for fullstack development of Icaro AI web app using **Dash** and **Plotly** UI and **Python** backend hosted in **Heroku**.

SOFTWARE DEVELOPER (CONTRACT)

Wolfram Research | February 2021 – February 2022 | Remote

- Sysadmin. Fullstack dev of portal for "Wolfram Summer Camp" (+40 users).
- Published paper "Turing Patterns in Networks" in Complex Systems Journal.

PROJECTS

ARXIV DOMAIN EXPERT LLM

GitHub Repository

- Developed custom fine-tuning datasets in **Hugging Face**, fine-tuned Llama-3.2-3B on **AWS SageMaker** and deployed with vector database **Qdrant**.

MASTER THESIS: NEURAL MODEL IDENTIFICATION WITH EKF

Ph.D. Esteban A. Hernandez-Vargas | Systems Medicine of Infectious Diseases

- Derived robust EKF weight update rule for RNNs, maintaining stability up to 20% signal-to-noise ratio in viral time-series modeling of SARS-Cov-2, HIV, etc...